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Session Roadmap

Part 1: Python Basics

-  Python Environment Setup
-  Variables & Data Types
-  Data Structures
-  Control Flow

Part 2: Data Manipulation

-  Pandas Fundamentals
-  NumPy Basics
-  Data Loading & Export

Part 3: Visualization

-  Matplotlib
-  Seaborn
-  Plotly

Part 4: Professional Techniques

-  Data Cleaning
-  Feature Engineering
-  Best Practices

Python Basics

Python Environment Setup

Recommended Tools

-  **IDE:** VS Code / PyCharm
-  **Jupyter Notebook** – Interactive
-  **Google Colab** – Cloud-based
-  **Anaconda** – Package Manager

Key Libraries

-  **NumPy** – Numerical computing
-  **Pandas** – Data manipulation
-  **Matplotlib** – Visualization
-  **Seaborn** – Statistical plots
-  **Scikit-learn** – Machine Learning

>_ Installation Commands

```
pip install numpy pandas  
matplotlib seaborn pip install  
jupyter scikit-learn
```

Variables & Data Types

Core Concepts

- **Variables** – Named storage
- **Data Types** – Type of data
- **Type Conversion** – Changing types

Common Data Types

- `int` – Integer numbers
- `float` – Decimal numbers
- `str` – Text strings
- `bool` – True/False
- `list` – Ordered collection
- `dict` – Key-value pairs

</> Code Example

```
1 # Integer
2 age = 25
3
4 # Float
5 price = 99.99
6
7 # String
8 name = "INEC Staff"
9
10 # Boolean
11 is_active = True
12
13 # List
14 voters = ["John", "Mary", "Peter"]
15
16 # Dictionary
17 results = { "LGA": "Abuja", "Votes": 1500 }
```

Data Structures

Lists

- Ordered, mutable collection
- Supports indexing & slicing

Tuples

- Ordered, immutable collection
- Faster than lists

Dictionaries

- Key-value pairs
- Fast lookups

</> Code Example

```
1 % Lists
2 results = [1200, 1500, 980, 2100]
3 results[0] % Access first element
4 results.append(1800) % Add item
5
6 % Tuples
7 coordinates = (6.5244, 3.3792)
8
9 % Dictionaries
10 election_data = {"state": "Lagos", "total_votes":
11                  25000, "turnout": 0.65 }
12 election_data["state"]
```

Control Flow

Conditionals

- if, elif, else
- Decision making

Loops

- for – Iterate over sequences
- while – Repeat while condition

</> Code Example

```
1 % If statement
2 turnout = 0.72
3 if turnout > 0.70:
4     print("High turnout")
5 elif turnout > 0.50:
6     print("Moderate turnout")
7 else:
8     print("Low turnout")
9
10 % For loop
11 for lga in ["Abuja", "Lagos", "Kano"]:
12     print(f"Processing: {lga}")
13
14 % While loop
15 i = 0
16 while i < 5:
17     print(i)
18     i += 1
```

Data Manipulation

NumPy Fundamentals

What is NumPy?

- Numerical Python
- N-dimensional arrays
- Mathematical functions
- Linear algebra operations

Key Features

- Fast array operations
- Broadcasting
- Vectorization
- Random number generation

</> Code Example

```
1 import numpy as np
2
3 % Create arrays
4 votes = np.array([1200, 1500, 980, 2100])
5
6 % Array operations
7 total_votes = np.sum(votes)
8 avg_votes = np.mean(votes)
9 std_votes = np.std(votes)
10
11 % Vectorized operations
12 turnout_rate = votes / 2000 * 100
13
14 % Random numbers
15 random_data = np.random.randn(100)
```

Pandas Fundamentals

What is Pandas?

- Data manipulation library
- **Series** – 1D labeled arrays
- **DataFrame** – 2D tabular data

Key Operations

- Reading/Writing data
- Data cleaning
- Aggregation
- Merging & Joining
- Grouping

</> Code Example

```
1 import pandas as pd
2
3 % Create DataFrame
4 data = {"LGA": ["Abuja", "Lagos", "Kano"], "Votes": [12000, 25000, 18000], "Turnout": [0.65, 0.72, 0.58]}
5 df = pd.DataFrame(data)
6
7 % Inspect data
8 df.head()
9 df.info()
10 df.describe()
11
12 % Filtering
13 high_turnout = df[df["Turnout"] > 0.70]
```

Loading Election Data

Data Sources

- CSV files
- Excel files
- JSON data
- Databases (SQL)
- Web APIs

Professional Practice

- Always check data types
- Handle missing values
- Validate data integrity

</> Code Example

```
1 % Read CSV
2 df = pd.read_csv("election_results.csv")
3
4 % Read Excel
5 df = pd.read_excel("election_data.xlsx",
6                     sheet_name="Results")
7
8 % Inspect data types
9 df.dtypes
10
11 % Handle missing values
12 df = df.dropna() % Remove missing
13 df = df.fillna(0) % Fill with 0
14
15 % Save processed data
16 df.to_csv("cleaned_results.csv")
```

Data Visualization

Matplotlib – The Foundation

Why Matplotlib?

- Industry standard
- Highly customizable
- Integration with Pandas
- Publication quality

Common Plots

- Line plots
- Bar charts
- Histograms
- Scatter plots
- Pie charts

</> Code Example

```
1 import matplotlib.pyplot as plt
2
3 % Line plot
4 years = [2019, 2020, 2021, 2022]
5 votes = [15000, 18000, 20000, 25000]
6
7 plt.plot(years, votes, marker="o", color="blue")
8 plt.title("Voter Turnout Over Years")
9 plt.xlabel("Year")
10 plt.ylabel("Votes")
11 plt.grid(True)
12 plt.show()
13
14 % Bar chart
15 plt.bar(years, votes, color="green")
```

Professional Visualization Techniques

Seaborn

- Statistical visualizations
- Built-in themes
- Advanced plots

Plotly

- Interactive plots
- Web-based
- Dashboard integration

</> Code Example

```
1 import seaborn as sns
2
3 % Set professional style
4 sns.set_style("whitegrid")
5
6 % Correlation heatmap
7 plt.figure(figsize=(10, 8))
8 sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
9
10 plt.title("Correlation Matrix")
11 plt.show()
12
13 % Pairplot
14 sns.pairplot(df, hue="Region")
15 plt.show()
16
17 % Histogram with KDE
18 sns.histplot(df["Turnout"], kde=True)
```

Professional Techniques

Data Cleaning Best Practices

Key Steps

- Check for duplicates
- Handle missing values
- Fix data types
- Remove outliers
- Standardize formats

Professional Tips

- Document all steps
- Keep raw data safe
- Use version control
- Validate results

</> Code Example

```
1 % Duplicate check
2 df.duplicated().sum()
3 df = df.drop_duplicates()
4
5 % Missing values
6 df.isnull().sum()
7 df = df.fillna(df.median())
8
9 % Outlier detection
10 Q1 = df["Votes"].quantile(0.25)
11 Q3 = df["Votes"].quantile(0.75)
12 IQR = Q3 - Q1
13 df = df[(df["Votes"] >= Q1 - 1.5*IQR) & (df["Votes"
    ""] <= Q3 + 1.5*IQR)]
```


Feature Engineering

What is Feature Engineering?

- Creating new features
- Transforming existing ones
- Improving model performance

Common Techniques

- Scaling/Normalization
- Encoding categories
- Creating interactions
- Date/Time features
- Aggregation

</> Code Example

```
1 from sklearn.preprocessing import StandardScaler
2
3 % Create new features
4 df["Turnout_percent"] = df["Votes"] / df["
    Registered"] * 100
5
6 % Encoding categorical
7 df["LGA_encoded"] = df["LGA"].astype('category').
    cat.codes
8
9 % Scaling numeric features
10 scaler = StandardScaler()
11 df["Votes_scaled"] = scaler.fit_transform(df[["
    Votes"]])
12
13 % Date features
14 df["Year"] = pd.to_datetime(df["Date"]).dt.year
15 df["Month"] = pd.to_datetime(df["Date"]).dt.month
```

Exploratory Data Analysis

Exploratory Data Analysis (EDA)

What is EDA?

- Understanding data
- Finding patterns
- Generating hypotheses
- Communicating insights

EDA Framework

- Summary statistics
- Missing data analysis
- Distribution analysis
- Correlation analysis
- Visualization

</> Code Example

```
1 % Summary statistics
2 df.describe()
3
4 % Data info
5 df.info()
6
7 % Distribution analysis
8 df["Votes"].value_counts().head(10)
9
10 % Correlation matrix
11 df.corr()
12
13 % Visual EDA
14 import seaborn as sns
15 sns.pairplot(df, hue="Region")
16 plt.show()
17
18 % Box plots
19 sns.boxplot(x="Region", y="Votes", data=df)
```

Best Practices

Python Best Practices

Code Quality

- Use meaningful variable names
- Write comments and docstrings
- Follow Python Enhancement Proposal (PEP) 8
- Use functions and modules

Professional Workflow

- Use virtual environments
- Version control with Git
- Write unit tests
- Document your work

</> Code Example

```
1 def clean_election_data(df):
2     """
3     Clean election results data.
4
5     Args:
6         df: Raw election dataframe
7     Returns:
8         Cleaned dataframe
9     """
10    # Remove duplicates
11    df = df.drop_duplicates()
12
13    # Handle missing values
14    df = df.fillna({"Votes": 0, "Turnout": df["
15                    Turnout"].median()})
16
17    return df
18
19    % Usage
20    clean_df = clean_election_data(raw_df)
```

Key Takeaways

Key Takeaways

What We Covered

1. **Python Basics** – Variables, data types, control flow
2. **Data Structures** – Lists, tuples, dictionaries
3. **NumPy** – Numerical computing
4. **Pandas** – Data manipulation
5. **Visualization** – Matplotlib, Seaborn
6. **Professional Techniques** – EDA, cleaning, feature engineering

💡 Next Steps

-  Practice with real datasets
-  Work with INEC election data
-  Continue learning

Resources

-  W3Schools Python Tutorial
-  Pandas Documentation
-  Kaggle Learn Python

Practical Exercise

Mini Project

1. Load an election dataset
2. Clean the data
3. Calculate key statistics
4. Create visualizations
5. Generate a report



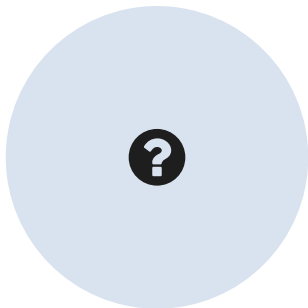
Sample Dataset

```
1 % Dataset: INEC Election Results
2 % Columns: LGA, State, Total_Votes,
3 %           Registered, Turnout, Region
4 % Task: Analyze and visualize turnout patterns
```

Time: 30 minutes — Work in teams of 3-4

Thank You

Questions & Discussion



"Mastering Python is the first step to mastering data science."